# Linear Algebra — Exercises

Operations modulo $n$. Groups and fields

**Exercise 1.** Let $n$ be a fixed positive integer and let $a$ and $b$ be any two integers. Recall that $a$ is congruent to $b$ modulo $n$ if $n$ divides $a - b$. Calculate: a) $17 \bmod 4$ b) $4 \bmod 17$ c) $(-2) \bmod 5$

**Exercise 2.** For integers $x$ and $y$ we write $x \equiv y \pmod{n}$ (read: $x$ is congruent to $y$ modulo $n$) if $n$ divides $x - y$. Solve the equations:

a) $2x \equiv 1 \pmod{7}$

b) $2x \equiv 1 \pmod{6}$

c) $5x \equiv 1 \pmod{9}$

d) $x^2 \equiv 3 \pmod{11}$

e) $2x + 3 \equiv 0 \pmod{5}$

f) $x + k \equiv 0 \pmod{n}$

**Exercise 3.** Prove that for all $n, k, p \in \mathbb{Z}$ we have the following:

a) $(n \bmod k) \bmod k = n \bmod k$

b) $(n + p) \bmod k = \big((n \bmod k) + (p \bmod k)\big) \bmod k$

c) $(np) \bmod k = \big((n \bmod k)\,(p \bmod k)\big) \bmod k$

**Exercise 4.** Recall that for $a, b \in \mathbb{N}$ the operations of addition and multiplication modulo $n$ are defined by $a +_n b = (a + b) \bmod n$ and $a \cdot_n b = (a \cdot b) \bmod n$. Prove that:

a) addition modulo $n$ is associative, i.e. $(a +_n b) +_n c = a +_n (b +_n c)$ for any $a, b, c \in \mathbb{N}$;

b) multiplication modulo $n$ is associative, i.e. $(a \cdot_n b) \cdot_n c = a \cdot_n (b \cdot_n c)$ for any $a, b, c \in \mathbb{N}$;

c) multiplication modulo $n$ is distributive with respect to addition modulo $n$, i.e. $(a +_n b) \cdot_n c = (a \cdot_n c) +_n (b \cdot_n c)$ for any $a, b, c \in \mathbb{N}$.

**Exercise 5.** State which of the following are groups: a) $(\mathbb{Z}_n, +_n)$ b) $(\mathbb{Z}_n \setminus \{0\}, \cdot_n)$ c) $(\{1, 2, 3, 4, 6, 12\}, \gcd)$

**Exercise 6.** State which of the following are groups: a) $(\{z \in \mathbb{C} \mid |z| = 1\}, +)$ b) $(\{z \in \mathbb{C} \mid |z| = 1\}, \cdot)$

**Exercise 7.** Show that for every $n$ all complex roots of 1 of order $n$ form a group under multiplication. Write down the table of the group of all complex roots of 1 of order 4.

**Exercise 8.** Show that $(2^X, \triangle)$ is a group, where $\triangle$ denotes the symmetric difference. Construct the group table of $(2^{\{a,b,c\}}, \triangle)$.

**Exercise 9.** Show that $(\mathbb{Z}_n \setminus \{0\}, \cdot_n)$ is a group if and only if $n$ is a prime number.

**Exercise 10.** Verify whether $(\mathbb{R}_+, \bullet)$ is a group, where $a \bullet b = a^2 b^2$.

**Exercise 11.** Let $p \in \mathbb{N}$ be a prime number. Show that $(\mathbb{Z}_p, +_p, \cdot_p)$ is a field.

**Exercise 12.** Determine which of the following structures are fields. Do this in two steps: first verify whether the set is a group with respect to the first operation, and then whether the set without the identity element of the first operation is a group with respect to the second operation. Here $2^X$ carries the operations $\cup$, $\cap$ and the symmetric difference $\triangle$, while $\mathbb{R}^{\mathbb{R}}$ denotes the set of all functions

$\mathbb{R} \to \mathbb{R}$, with pointwise addition $+$ and composition $\circ$. a) $(2^X, \cup, \cap)$ b) $(\mathbb{R}^{\mathbb{R}}, +, \circ)$ c) $(2^X, \cap, \cup)$ d) $(\mathbb{R}^{\mathbb{R}}, \circ, +)$ e) $(2^X, \cap, \triangle)$ f) $(2^X, \triangle, \cap)$ g) $(2^X, \cup, \triangle)$ h) $(2^X, \triangle, \cup)$

**Exercise 13.** Verify whether $(\mathbb{R}, \oplus, \odot)$ is a field, where $a \oplus b = a + b + 1$ and $a \odot b = ab + a + b$.